Suggested problems: 1, 3, 7, 8, 10

Problem 1: Prove that if $A \subset B$, then $B^c \subset A^c$.

Problem 2: Suppose that a number x is to be selected from the real line S, and let A and B be the following events:

$$A = \{x : 1 \le x \le 3\},$$
$$B = \{x : 3 \le x \le 7\}.$$

Describe each of the following events.

(a) A^c

(b) $A \cup B$

(c) $A^c \cap B^c$

Problem 3: Suppose that a coin is tossed three times with an observer watching. Identify an appropriate sample space for this experiment, and then discuss how the collections of events change as the observer sees the result of the first flip, then the second, and finally the third.